

CLINICAL STUDY SERIES

PHARMACOLOGY CHEAT SHEETS: CARDIOVASCULAR & RENAL

By Drug Family · Landscape Tables · 10 Printable Pages

HOW TO USE

- Print at 100% / "Actual Size" on US Letter paper.
- Work from memory first, then check your notes and textbook.
- Reprint blank copies for each drug class or exam unit.

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BETA BLOCKERS

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Metoprolol BOXED WARNING	Beta Blocker (Selective)	HTN, angina, heart failure, MI	Blocks beta-1 receptors, decreasing HR and contractility	Bradycardia, hypotension, fatigue, masks hypoglycemia	Check HR/BP before giving. Do not stop abruptly.
Atenolol BOXED WARNING	Beta Blocker (Selective)	HTN, Angina, Acute MI	Selectively blocks beta-1 adrenergic receptors	Fatigue, bradycardia, dizziness	Hold if HR < 60 bpm. Taper when discontinuing.
Propranolol BOXED WARNING	Beta Blocker (Non-selective)	HTN, migraines, performance anxiety	Blocks Beta-1 (heart) and Beta-2 (lungs) receptors	Bradycardia, bronchospasm, cold extremities	Strictly contraindicated in asthma/COPD patients.
Carvedilol	Beta/Alpha Blocker	Heart failure, HTN	Non-selective beta and alpha-1 blockade (causes vasodilation)	Bradycardia, orthostatic hypotension, fatigue	Take with food to minimize orthostatic hypotension.
Labetalol	Beta/Alpha Blocker	Severe HTN, hypertensive crisis	Non-selective beta and alpha-1 blockade	Orthostatic hypotension, dizziness, scalp tingling	Keep patient supine during IV administration.
Esmolol	Beta Blocker (Selective)	SVT, Atrial Fibrillation (Rapid rate control)	Ultra-short-acting beta-1 blocker	Profound hypotension, diaphoresis	Given via continuous IV infusion. Very rapid onset/offset.

RAAS BLOCKERS (ACE / ARB / ARNI)

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Lisinopril BOXED WARNING	ACE Inhibitor	HTN, heart failure, post-MI	Blocks conversion of Angiotensin I to Angiotensin II	Dry cough, hyperkalemia, angioedema	Monitor BP and K+. Teratogenic (avoid in pregnancy).
Enalapril BOXED WARNING	ACE Inhibitor	HTN, Heart Failure	Inhibits ACE, reducing Angiotensin II and Aldosterone	Dry nagging cough, angioedema, hyperkalemia	Can be given IV (Enalaprilat). Monitor renal function.
Captopril BOXED WARNING	ACE Inhibitor	HTN, Diabetic Nephropathy	Blocks conversion of Angiotensin I to II	Cough, loss of taste, neutropenia	Shortest half-life of ACEs. Take 1hr before meals.
Losartan BOXED WARNING	Angiotensin II Receptor Blocker (ARB)	HTN, diabetic nephropathy	Blocks Angiotensin II directly at the receptor site	Dizziness, hyperkalemia, hypotension	No dry cough (alternative for ACE intolerance). Teratogenic.
Valsartan BOXED WARNING	Angiotensin II Receptor Blocker (ARB)	HTN, Heart Failure	Blocks Angiotensin II receptors	Dizziness, hyperkalemia	Often combined with Sacubitril (Entresto) for heart failure.
Irbesartan BOXED WARNING	Angiotensin II Receptor Blocker (ARB)	HTN, Diabetic Nephropathy	Blocks Angiotensin II receptors	Orthostatic hypotension, hyperkalemia	Monitor BUN/Creatinine and Potassium regularly.
Sacubitril/Valsartan <i>Entresto</i> BOXED WARNING	ARNI	Heart Failure with Reduced Ejection Fraction (HFrEF)	Neprilysin inhibitor (increases natriuretic peptides) + ARB	Hypotension, hyperkalemia, angioedema	Must have a 36-hour washout period if switching from an ACE inhibitor.

CALCIUM CHANNEL BLOCKERS

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Amlodipine	Calcium Channel Blocker (DHP)	HTN, chronic stable angina	Inhibits calcium influx into vascular smooth muscle	Peripheral edema, dizziness, flushing	Educate patient to report severe swelling in legs/ankles.
Nifedipine	Calcium Channel Blocker (DHP)	HTN, angina, preterm labor	Inhibits calcium influx into vascular smooth muscle	Reflex tachycardia, peripheral edema, gingival hyperplasia	Do not crush extended-release tablets.
Diltiazem	Calcium Channel Blocker (Non-DHP)	AFib rate control, angina, HTN	Inhibits calcium influx, slowing AV node conduction	Bradycardia, peripheral edema, constipation	Monitor HR and BP. Used IV for rapid rate control in AFib.
Verapamil	Calcium Channel Blocker (Non-DHP)	Angina, SVT, HTN, migraines	Inhibits calcium influx, deeply depressing SA/AV nodes	Severe constipation, bradycardia, hypotension	Highest risk of constipation among CCBs. Increase fluids/fiber.

LIPID-LOWERING DRUGS

PHARMACOLOGY CHEAT SHEET

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Atorvastatin	Statin / Antilipemic	Hyperlipidemia, CV risk reduction	Inhibits HMG-CoA reductase, decreasing cholesterol	Myopathy, rhabdomyolysis, hepatotoxicity	Can be taken any time of day. Avoid large amounts of grapefruit juice.
Simvastatin	Statin / Antilipemic	Hyperlipidemia, CV event prevention	Inhibits HMG-CoA reductase	Muscle pain, elevated LFTs, rhabdomyolysis	Take in the evening. Report unexplained muscle aches.
Rosuvastatin	Statin / Antilipemic	Hyperlipidemia	Inhibits HMG-CoA reductase	Myopathy, elevated LFTs	High intensity statin. Monitor CPK if muscle pain occurs.
Ezetimibe	Cholesterol Absorption Inhibitor	Hyperlipidemia	Inhibits absorption of cholesterol in the small intestine	Hepatitis, myopathy, diarrhea	Often combined with a statin for synergistic effect.
Cholestyramine	Bile Acid Sequestrant	Hyperlipidemia	Binds bile acids in the GI tract, increasing cholesterol excretion	Severe constipation, bloating, belching	Interferes with absorption of many other drugs. Space dosing out.
Gemfibrozil	Fibrate	Hypertriglyceridemia	Decreases hepatic triglyceride secretion, increases VLDL clearance	Gallstones, myopathy, hepatotoxicity	High risk of myopathy if given concurrently with a statin.
Evolocumab	PCSK9 Inhibitor	Familial hypercholesterolemia, clinical ASCVD	Monoclonal antibody that increases LDL receptor recycling	Injection site reactions, nasopharyngitis	Administered via subcutaneous injection every 2-4 weeks.

DIURETICS (1/2)

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Furosemide	Loop Diuretic	Heart failure, severe edema, HTN	Inhibits Na/Cl reabsorption in ascending Loop of Henle	Hypokalemia, ototoxicity, profound polyuria	Push IV slowly (20mg/min) to avoid hearing loss. Monitor K+.
Bumetanide	Loop Diuretic	Heart Failure, Edema	Inhibits Na/Cl reabsorption in Loop of Henle	Profound hypokalemia, ototoxicity	40x more potent than furosemide. Close electrolyte monitoring required.
Torsemide	Loop Diuretic	Edema associated with heart failure, renal disease	Inhibits Na/Cl reabsorption in ascending Loop of Henle	Hypokalemia, hypomagnesemia, dehydration	Longer half-life and better oral bioavailability than furosemide.
Hydrochlorothiazide	Thiazide Diuretic	HTN, peripheral edema	Inhibits Na/Cl reabsorption in distal convoluted tubule	Hypokalemia, hyponatremia, hyperuricemia, hyperglycemia	Take in the morning. Not effective if severe renal impairment.
Chlorthalidone	Thiazide-like Diuretic	HTN, edema	Inhibits sodium reabsorption in the distal tubule	Hypokalemia, hyperuricemia, dizziness	Longer duration of action than HCTZ. Preferred for HTN.
Spironolactone	Potassium-Sparing Diuretic	Heart failure, ascites, HTN	Aldosterone antagonist in distal tubule	Hyperkalemia, gynecomastia, amenorrhea	Avoid potassium supplements and salt substitutes.
Eplerenone	Potassium-Sparing Diuretic	Heart failure post-MI, HTN	Selective aldosterone receptor antagonist	Hyperkalemia, dizziness, hypertriglyceridemia	Less risk of gynecomastia compared to spironolactone.

DIURETICS (2/2)

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Acetazolamide	Carbonic Anhydrase Inhibitor	Glaucoma, altitude sickness, edema	Inhibits carbonic anhydrase, decreasing aqueous humor	Metabolic acidosis, hypokalemia, paresthesia	Contraindicated in patients with severe sulfa allergies.
Mannitol	Osmotic Diuretic	Increased Intracranial Pressure (ICP), Cerebral Edema	Increases osmotic pressure of glomerular filtrate	Fluid/electrolyte imbalances, pulmonary edema	Must use a filter needle/tubing. Inspect for crystals before giving.

OTHER BP DRUGS & VASODILATORS

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Clonidine	Alpha-2 Agonist	HTN, ADHD, Opioid withdrawal	Stimulates central alpha-2 receptors, decreasing sympathetic outflow	Dry mouth, severe sedation, rebound hypertension	Do not stop abruptly (hypertensive crisis risk).
Methyldopa	Alpha-2 Agonist	Hypertension in Pregnancy	Central alpha-2 agonist, reducing sympathetic tone	Sedation, positive Coombs test, hemolytic anemia	Drug of choice for pregnancy-induced hypertension.
Doxazosin	Alpha-1 Blocker	HTN, BPH	Blocks peripheral alpha-1 receptors causing vasodilation	First-dose orthostatic hypotension, dizziness	Give first dose at bedtime to prevent syncope (fainting).
Hydralazine	Direct Vasodilator	Severe HTN, Heart Failure	Directly relaxes arteriolar smooth muscle	Reflex tachycardia, fluid retention, Lupus-like syndrome	Often given with a beta-blocker to prevent reflex tachycardia.
Minoxidil BOXED WARNING	Direct Vasodilator	Severe, refractory HTN (also alopecia)	Opens potassium channels in smooth muscle, hyperpolarizing cell	Severe fluid retention, hypertrichosis (hair growth), pericardial effusion	Extremely potent. Only used when all other HTN drugs fail.
Nitroglycerin	Nitrate / Vasodilator	Angina (chest pain)	Converts to nitric oxide, causing rapid venous dilation	Severe headache, orthostatic hypotension, reflex tachycardia	Keep in dark glass bottle. Call 911 if pain persists after 1 dose.
Isosorbide Mononitrate	Nitrate / Vasodilator	Angina prophylaxis	Produces nitric oxide to dilate veins and coronary arteries	Headache, hypotension, dizziness	Requires a 10-14 hour nitrate-free period daily to prevent tolerance.

ANTIARRHYTHMICS & DIGOXIN

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Amiodarone BOXED WARNING HIGH-ALERT	Class III Antiarrhythmic	V-Fib, V-Tach, AFib	Prolongs action potential and refractory period (K+ block)	Pulmonary toxicity, thyroid dysfunction, blue-gray skin, corneal deposits	Extremely long half-life. Requires baseline PFTs, TFTs, and LFTs.
Sotalol BOXED WARNING HIGH-ALERT	Class III Antiarrhythmic	Ventricular arrhythmias, AFib maintenance	Non-selective beta blocker with Class III (K+ block) properties	Bradycardia, QT prolongation, Torsades de Pointes	Must be initiated in a hospital with continuous ECG monitoring (3 days).
Flecainide BOXED WARNING	Class IC Antiarrhythmic	SVT, AFib, life-threatening ventricular arrhythmias	Potent sodium channel blockade, significantly slowing conduction	Proarrhythmic (can cause lethal arrhythmias), dizziness	Contraindicated in patients with structural heart disease or prior MI.
Digoxin HIGH-ALERT	Cardiac Glycoside	Heart Failure, Atrial Fibrillation	Inhibits Na/K ATPase pump, increasing intracellular calcium (positive inotrope)	Anorexia, N/V, visual halos (yellow/green), bradycardia	Narrow therapeutic range (0.5-2.0 ng/mL). Hypokalemia increases toxicity risk.

ANTICOAGULANTS

PHARMACOLOGY CHEAT SHEET

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Warfarin BOXED WARNING HIGH-ALERT	Anticoagulant (Vitamin K Antagonist)	DVT, PE, AFib, Valve replacement	Inhibits synthesis of Vitamin K-dependent clotting factors (II, VII, IX, X)	Bleeding, bruising, tissue necrosis	Monitor PT/INR (Target 2-3). Maintain consistent Vit K diet. Antidote: Vitamin K.
Heparin HIGH-ALERT	Anticoagulant (Indirect Thrombin Inhibitor)	Acute DVT, PE, MI, AFib	Binds to antithrombin III, inactivating thrombin and factor Xa	Bleeding, Heparin-Induced Thrombocytopenia (HIT)	Monitor aPTT. High-alert medication. Antidote: Protamine Sulfate.
Enoxaparin <i>Lovenox</i> BOXED WARNING HIGH-ALERT	Low Molecular Weight Heparin (LMWH)	DVT prophylaxis, acute DVT/PE, ACS	Potentiates antithrombin III, preferentially inactivating factor Xa	Bleeding, thrombocytopenia, injection site hematoma	Give deep SQ in the abdomen. Do not expel the air bubble in the syringe.
Apixaban <i>Eliquis</i> BOXED WARNING HIGH-ALERT	DOAC / Factor Xa Inhibitor	DVT/PE treatment, AFib stroke prevention	Directly and reversibly inhibits Factor Xa	Bleeding, bruising, GI hemorrhage	No routine blood monitoring required. Antidote: Andexanet alfa.
Rivaroxaban <i>Xarelto</i> BOXED WARNING HIGH-ALERT	DOAC / Factor Xa Inhibitor	DVT/PE treatment, AFib	Directly inhibits Factor Xa	Bleeding, bruising	Take with evening meal (improves absorption). No routine INR needed.
Dabigatran <i>Pradaxa</i> BOXED WARNING HIGH-ALERT	Direct Thrombin Inhibitor	AFib stroke prevention, DVT/PE	Direct, reversible, competitive inhibition of thrombin	Bleeding, dyspepsia, gastritis	Keep in original moisture-proof container. Antidote: Idarucizumab.

ANTIPLATELETS & THROMBOLYTICS

1 - Cardiovascular & Renal

DRUG	CLASS	USES	MECHANISM	SIDE EFFECTS	NURSING NOTES
Clopidogrel <i>Plavix</i> BOXED WARNING	Antiplatelet Agent	Prevent MI/Stroke, post-stent placement	Irreversibly inhibits platelet aggregation (ADP receptor)	Bleeding, bruising, TTP	Discontinue 5-7 days before planned surgery.
Ticagrelor <i>Brilinta</i> BOXED WARNING	Antiplatelet Agent	Acute Coronary Syndrome (ACS), post-stent	Reversibly inhibits the platelet P2Y12 ADP receptor	Bleeding, dyspnea (shortness of breath), bradycardia	Efficacy is decreased by daily aspirin doses >100mg.
Alteplase (tPA) HIGH-ALERT	Thrombolytic	Acute Ischemic Stroke, massive PE, STEMI	Binds to fibrin, converting plasminogen to plasmin to dissolve clots	Intracranial hemorrhage, severe systemic bleeding	Must be given within 3-4.5 hours of symptom onset for stroke. Strict exclusion criteria.

EDUCATIONAL USE ONLY

NOT A CLINICAL REFERENCE

This study aid is a learning and exam-review tool for nursing and pharmacy students. It must not be used to make patient-care decisions, including dosing, administration, or monitoring.

ALWAYS VERIFY

Drug information, boxed (black box) warnings, and high-alert designations change over time. Before applying anything in practice, check the current FDA-approved labeling, your facility's policies, and an up-to-date drug reference. Normal lab ranges vary between laboratories; use the ranges your program or facility provides.

RECOMMENDED REFERENCES FOR VERIFICATION

- FDA drug labels: DailyMed (dailymed.nlm.nih.gov)
- ISMP List of High-Alert Medications ([ismp.org](https://www.ismp.org))
- A current nursing drug handbook or your program's required drug reference
- Your course textbook and instructor guidance

ACCURACY

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